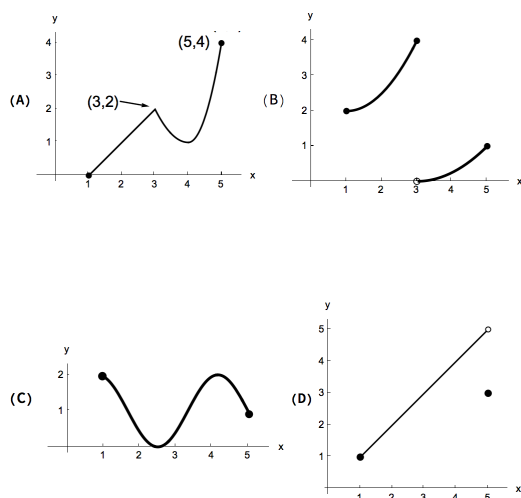


Problem 6

Problem 6

Problem 1 Given the four functions on the interval $[1, 5]$, answer the questions below.



- List the functions that satisfy the hypothesis of the Extreme Value Theorem on $[1, 5]$.
- For each of the functions, determine if they have a global maximum, a global minimum, both, or neither.
- List the functions that satisfy the hypothesis of the Mean Value Theorem on $[1, 5]$.
- List the function (or functions) for which there exists a point c in $(1, 5)$ such that

$$f'(c) = \frac{f(5) - f(1)}{5 - 1}$$